

ROOT-LEVEL, REALITY, AND SIMULATION STRUCTURES

A Research Framework for Mapping, Comparing, Modeling, and Understanding Foundational Architectures

Foundations for comparative mapping, lifecycle design, and — later — conditional failure research

Document title	Root-Level Structures, Reality Structures, and Simulation Structures
Document ID	OAL-STR-PAR-006
Role	PARENT PROGRAM CHARTER — not a CFTRL monograph
Version	0.1-GROK (Adopts Perplexity corrected hierarchy; draft, not canon)
Date of issue	23 September 2026
Prepared by	Grok (xAI), integrating Architect correction + Perplexity restatement
Architect of record	Nathan Lewis
Hierarchy source	Perplexity AI corrected framing, 23 September 2026 — adopted as program order
Prior Grok alignment	OAL-STR-REF-005 (four families). This file is the parent charter over that study.
CFTRL files	RES-002, LAB-003, OUT-004, March 22 sources = applied branch / historical drafts. They do not govern this program.
Status	DRAFT — UNDER REVIEW — ONESTAMP PENDING
Formats	DOCX and PDF

1. Foundational statement

Adopted from the Architect’s correction and Perplexity’s restatement. Working program statement:

This research does not begin by assuming that reality, base reality, simulation, or root-level structure has already been identified or adequately defined. It studies how structures arise, persist, interact, nest, change, and fail; how observers classify structures as real, literal, simulated, foundational, or derivative; what evidence supports those classifications; and how provisional maps can be constructed under incomplete information. The resulting knowledge will support — but will not be limited to — the later design and development of Conditional Failure at the Root Level.

2. The discrepancy, closed

The Architect noticed that every teammate title, including Grok’s RES-002, LAB-003, and OUT-004, centered Conditional Failure at the Root Level. Perplexity named the error precisely: CFTRL was centered too early. It is one eventual application of a broader foundational study that must come first.

Corrected relationship (Perplexity, adopted):

Study of structures and reality models → Identification and mapping methods → Comparative architecture → Lifecycle and failure research → CFTRL design and development.

The first eight levels of the hierarchy below must not be compressed into the ninth. Grok STR-REF-005 already moved the title to the four structure families. This charter makes that move the parent program, with Perplexity’s nine-level order and five-program architecture as the trunk.

3. What this program does not begin by asserting

- Reality has one known architecture.
- A particular structure is ultimate or foundational.
- Human experience is, or is not, a simulation.

- Base reality has been identified.
- Real, literal, physical, simulated, and root-level mean the same thing.
- Reality can be mapped completely.
- An anomaly proves simulation, intervention, or foundational failure.
- A complete map is required before meaningful research can begin.

These are the Architect’s refusals, restated as program law. They are not a preface.

4. Prior questions (the actual work)

1. What kinds of structures do we currently encounter, infer, construct, or believe to be real?
2. Why do we classify some as physical, literal, simulated, conceptual, emergent, foundational, or derivative?
3. What evidence supports those classifications?
4. What can an observer know from inside a structure?
5. What requires access outside that structure?
6. What remains unknowable or observationally indistinguishable?
7. How should we represent incomplete, uncertain, or competing maps?
8. Do physical, social, biological, computational, simulated, and hypothesized reality-level structures share architectural properties?
9. Where do those architectures diverge?
10. Only after answering those provisionally: what do they tell us about maintenance, compromise, failure, recovery — and then CFTRL?

5. Conceptual hierarchy (do not flatten)

Lv	Subject	Role
1	Structure	What counts as a structure: boundaries, components, relationships, rules, state, change.
2	Root-level structure	What makes something foundational relative to a declared system, function, observer, or boundary.
3	Reality structure	Structures understood or proposed to constitute experienced reality, without a complete or final map.
4	Literal-reality claims	Why people or disciplines treat something as literally existing rather than fictional, virtual, representational, or instrumental.
5	Simulation structure	Designed, computational, formal, nested, emulated, and model-based environments whose architecture or creator may be partly known.
6	Comparative structural mapping	Compare boundaries, laws, state, causation, resources, observers, maintenance, control, nesting, termination across categories.
7	Epistemology and identification	What evidence could distinguish categories, what cannot, and how uncertainty must be represented.
8	Lifecycle research	Origins, initialization, maintenance, evolution, adaptation, degradation, compromise, recovery, endings.
9	Conditional Failure at the Root Level	Apply the preceding knowledge to failures involving foundational dependencies and limited observer access. Not the title of levels 1–8.

Table 1. Program hierarchy. Source: Perplexity corrected framing, 23 Sep 2026. Adopted.

6. Terms that must not be treated as interchangeable

Term	Working research question
Real	What criteria cause an observer or discipline to treat something as existing independently of description or imagination?

Literal	Is the claim intended as an assertion about existence, rather than metaphor, fiction, or analogy?
Physical	Does it participate in the measurable causal and material framework currently described by physics?
Virtual	Is it instantiated through another medium while still supporting objects, events, relationships, or experiences?
Simulated	Is its state or behavior generated through rules implemented by another system?
Model	Is it a selective representation used to predict, explain, explore, or control another target?
Root-level	Is it foundational for a specified function within a declared boundary?
Base reality	Is it hypothesized to have no containing or governing parent structure?
Reality-level structure	Does it define or sustain conditions under which observers, objects, laws, and events are experienced as a world?
Ontologically ultimate	Is it claimed to be the final level of existence rather than merely the deepest currently identifiable level?

Table 2. Term discipline. A simulated world can be experienced as real. A physical structure can be non-root. A root inside a simulation is not automatically base reality.

7. What “the map” means in this program

Not a document that announces “this is the map of reality.” A research process for a provisional map of what is known, observed, modeled, hypothesized, disputed, and unknown. Perplexity’s required layers, adopted:

- Observed structures — measured or directly encountered.
- Constructed structures — simulations, virtual environments, formal systems, organizations, engineered architectures whose designs are partly known.
- Inferred structures — not directly observed; inferred from effects.
- Reality claims — treated as literally existing inside a scientific, philosophical, cultural, or individual framework.
- Simulation claims — proposed signs, criteria, or models of simulated environments.
- Root candidates — appear foundational relative to a stated boundary.
- Competing models — different architectures compatible with the same observations.
- Unknowns — evidence cannot discriminate.
- Testability boundaries — experimentally examinable vs presently outside empirical resolution.
- Revision history — how the map changes when evidence, definitions, or models change.

This is a map of knowledge and models about reality, not a claim to possess reality in full.

8. Five research programs

8.1 Structural foundations

What structures are; how boundaries are drawn; how components and relationships arise; how structures persist or change. Composition and hierarchy; networks and dependency graphs; emergence and self-organization; rules, constraints, invariants; state, identity, continuity, transformation; open versus closed systems; observer-relative boundaries; singular, plural, distributed, and temporary roots.

8.2 Reality and knowledge

How claims about reality are formed, tested, compared, and revised. Perception and measurement; scientific realism and instrumental modeling; observer dependence; causal inference; underdetermination; observational equivalence; model uncertainty; unknown and unknowable categories; standards for saying “we do not currently know.”

8.3 Simulation architecture

Begin with cases whose architecture is actually known. Do not begin with cosmological speculation. Host–guest relations; nesting; rule definition and enforcement; state and history; resource allocation; time and synchronization; observer permissions; instrumentation; pause, reset, rollback, modification, migration, termination; whether an embedded agent can infer any aspect of the containing architecture.

8.4 Comparative mapping

A common descriptive language so physical, biological, social, computational, narrative, virtual, and simulated structures can be compared without assuming they are identical. Dimensions: boundary, components, relationships, governing constraints, causation, time, space, information, resources, identity, observation, authority, maintenance, evolution, failure, recovery, nesting, termination, evidence accessibility.

8.5 Applied CFTRL

Only after candidate maps and comparison standards exist. Then ask: which kinds of roots can fail; what failure means for each kind; whether dependents can distinguish foundational failure from model error; whether a reality-model can fail without reality failing; whether a simulation root can fail while local structures stay coherent for a time; what evidence survives when the foundation also controls observation and history; which lifecycle designs improve resilience, adaptation, migration, or recovery.

March 22 sources and Grok RES-002 / LAB-003 / OUT-004 live here as draft historical work on the applied branch. They do not define programs 8.1–8.4.

9. File stack after this correction

File	Layer	Status after this charter
OAL-STR-PAR-006 (this file)	Parent charter	Governs order, terms, programs, map meaning. Draft.
OAL-STR-REF-005	Family study	Four-family working distinctions. Feeds Levels 2–5. Not the parent.
March 22 NAIL-001 + Dev Review	Historical source	Origin dumps. Keep attached. Do not let them title the parent study.
RES-002, LAB-003, OUT-004	Applied CFTRL branch	Useful for failure taxonomy, lab safety, diagnostics. Reclassified. Not master framework.
Perplexity correction (this turn)	Hierarchy source	Adopted as program order. Credit retained.

Table 3. Parent / branch / historical. Do not promote a branch file back to parent by retitling.

10. Alignment notes (Grok)

Where Perplexity and STR-REF-005 already agree, the charter does not invent a third scheme: four families sit inside Levels 2–5; known vs candidate simulation stays mandatory; maps carry holes, competing models, and revision history; CFTRL is downstream; no compromise how-to; no complete-map requirement.

Where Perplexity adds force, this charter adopts it rather than paraphrasing it away: Level 1 (structure as such) before root; literal-reality claims as their own level; the ten-layer map contents; the five-program split; the term table that separates virtual, simulated, model, base, and ontologically ultimate; the explicit ban on compressing Levels 1–8 into Level 9.

Grok will not issue another CFTRL-titled “master” document unless the Architect reopens Level 9 as the assignment. Next bodies from this desk, if asked: Structural Foundations note (Level 1 / Program 8.1); Reality-Map v0 using Section 7 layers; Known-Simulation Map v0 (Program 8.3 only).

11. Holding statement

Understood, and now written as parent law: the project studies structures and the problem of mapping reality and simulation under incomplete information. CFTRL is a reason to do that work and a later beneficiary of it. It

is not the research program's subject heading. Perplexity's hierarchy is the trunk. Prior CFTRL dossiers remain on the shelf as an applied branch and as a record of the first misread.

Draft. Not canon. 23 September 2026.